

Manipal Institute of Technology, MAHE



SOFTWARE ENGINEERING LAB
(CSE 3141)

PORTKEY
Food Delivery System

Project Report

Submitted By:
Princita Zina Miranda (C-17)
Rithik Ramesh (C-11)
Abhay Nair (C-28)

- **Identify and design the suitable SDLC model for your project**

Criteria	Why Agile Fits Our Project
Incremental Delivery	We are building multiple modules (ordering, delivery, chatbot, payment) that can be released in phases.
Dynamic Requirements	Features like chatbot, delivery updates, etc., evolve with user feedback.
Frequent Testing	Agile encourages early and regular testing critical in UI-heavy and chatbot systems.
Team Collaboration	With 3 members, Agile enables daily/weekly sync and role switching.
User-Centered Design	Agile supports iterative improvements based on user experience with the UI and chatbot.

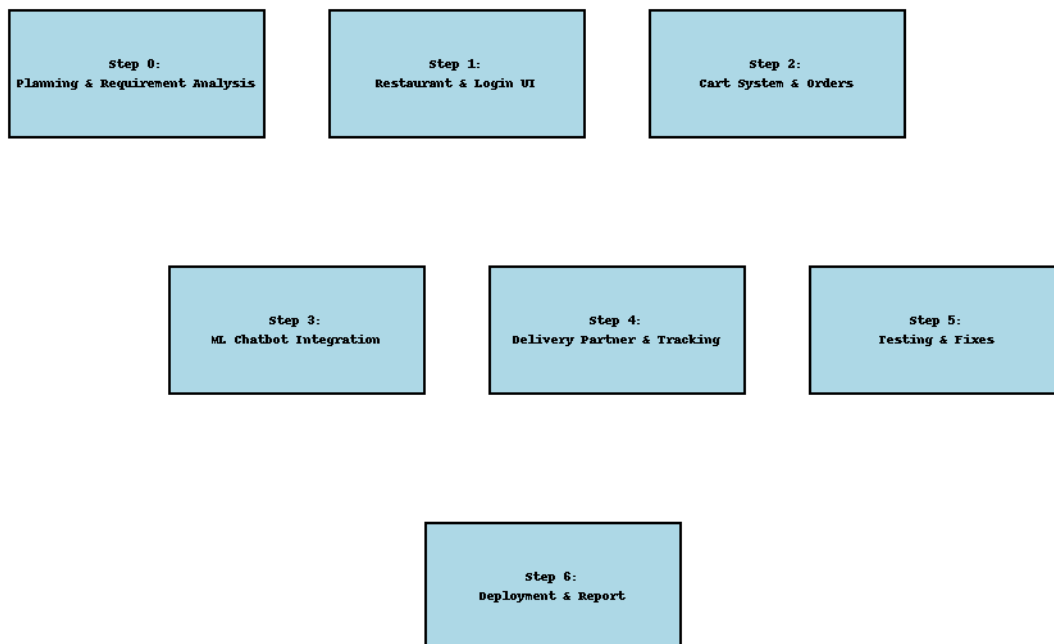
- **Justify the same**

- Step 0: Planning

- Define core modules: User authentication, restaurant listing, order placement, chatbot, delivery.
- Assign roles to group members.
- Select technologies: HTML/CSS, MySQL, Python (for ML).
- Break requirements into feature goals, like:
 - “Allow users to browse menu.”
 - “Enable chatbot to track orders.”

- Step 1: Restaurant Listing & User Login
 - Design restaurant menu interface.
 - Build basic HTML/CSS UI with Javascript.
 - Develop login & registration system.
 - Setup database schema for users and restaurants.
- Step 2: Cart & Order Placement System
 - Design cart UI and backend logic.
 - Implement order placement and confirmation.
 - Create backend APIs to handle menu, cart, and orders.
 - Store orders in the database.
- Step 3: ML Chatbot Integration
 - Design chatbot interface.
 - Train ML model.
 - Build an API to handle user queries.
 - Connect chatbot frontend to backend.
- Step 4: Delivery Partner Module & Tracking
 - Create delivery partner table and UI.
 - Simulate order tracking (e.g., “Order accepted”, “Out for delivery”, “Delivered”).
 - Add status update logic from backend.

- Step 5: Final Testing & Debugging
 - Unit testing (e.g., login, order).
 - Integration testing (UI + API).
 - Chatbot testing.
 - Visual/UI testing on different screen sizes.
- Step 6: Deployment & Documentation
 - Deploy chatbot API.
 - Host frontend.
 - Create user demo video (optional if time permits).
 - Finalize documentation, screenshots, and diagrams.
- **Give the model diagram**



- Explain every phase in detail based on your project

Step	Focus Modules	Outcome
Step 0	Planning & Requirement Analysis	Tools and responsibilities set
Step 1	Restaurant & Login UI	Users can log in and browse restaurants
Step 2	Cart System & Orders	Place orders with backend integration
Step 3	ML Chatbot Integration	Chatbot answers FAQs and tracks orders
Step 4	Delivery Partner & Tracking	Delivery logic and mock tracking enabled
Step 5	Testing & Fixes	Bugs resolved, chatbot and UI tested
Step 6	Deployment & Report	Project deployed and documented for grading